

An aerial night view of a city, likely Shanghai, with a network overlay of glowing nodes and connecting lines. The city lights are visible, and the network lines are white and curved, connecting various points across the city. The sky is a deep blue, and the city lights are warm yellow and orange.

Wireless Networks

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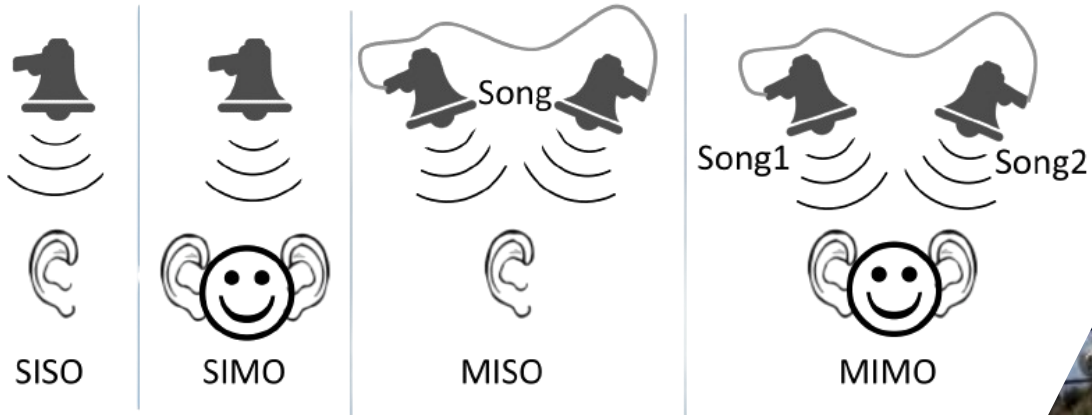
Diversity Techniques

Lecture 6



Contents

- ✓ Diversity in Wireless Communication
- ✓ Single input Single output (SISO)
- ✓ Single input Multiple output (SIMO)
- ✓ Multiple input Single output (MISO)
- ✓ Multiple Input Multiple Output (MIMO)



Diversity in Wireless Communication

Recap of Wireless Communication Channel Characteristics

■ Channel Characteristics & Performance

- Wireless system performance depends on the transmitter-receiver channel conditions.
- Key terminologies: **coherence bandwidth** and **coherence time**.

■ Coherence Bandwidth (B_c)

- Frequency range where the channel response is flat.
- Inversely proportional to the delay spread of the channel.
- If signal bandwidth $B_s \leq B_c$ → No distortion (flat fading).
- If signal bandwidth $B_s > B_c$ → Distorted signal (selective fading).

■ Coherence Time (T_c)

- Approximate time duration for which the channel remains constant.
- Inversely proportional to the Doppler frequency shift (B_d).
- If coherence time (T_c) $\geq$ symbol time → Slow fading.
- If coherence time (T_c) $<$ symbol time → Fast fading.

Diversity

- ▣ If you toss a coin, there is a 50% (or 1/2) chance of landing on **heads**.

$$\underbrace{\{H\}, \{T\}}_{1 \text{ out of } 2}$$



- ▣ If you toss two coins, there is a 75% (or 3/4) chance of getting at least one head.

$$\underbrace{\{H, H\}, \{H, T\}, \{T, H\}, \{T, T\}}_{3 \text{ out of } 4}$$

- ▣ For four coins, as shown in the figure above, there are $2^4=16$ possible outcomes, and the probability of getting at least one head increases to approximately 94% (or 15/16).

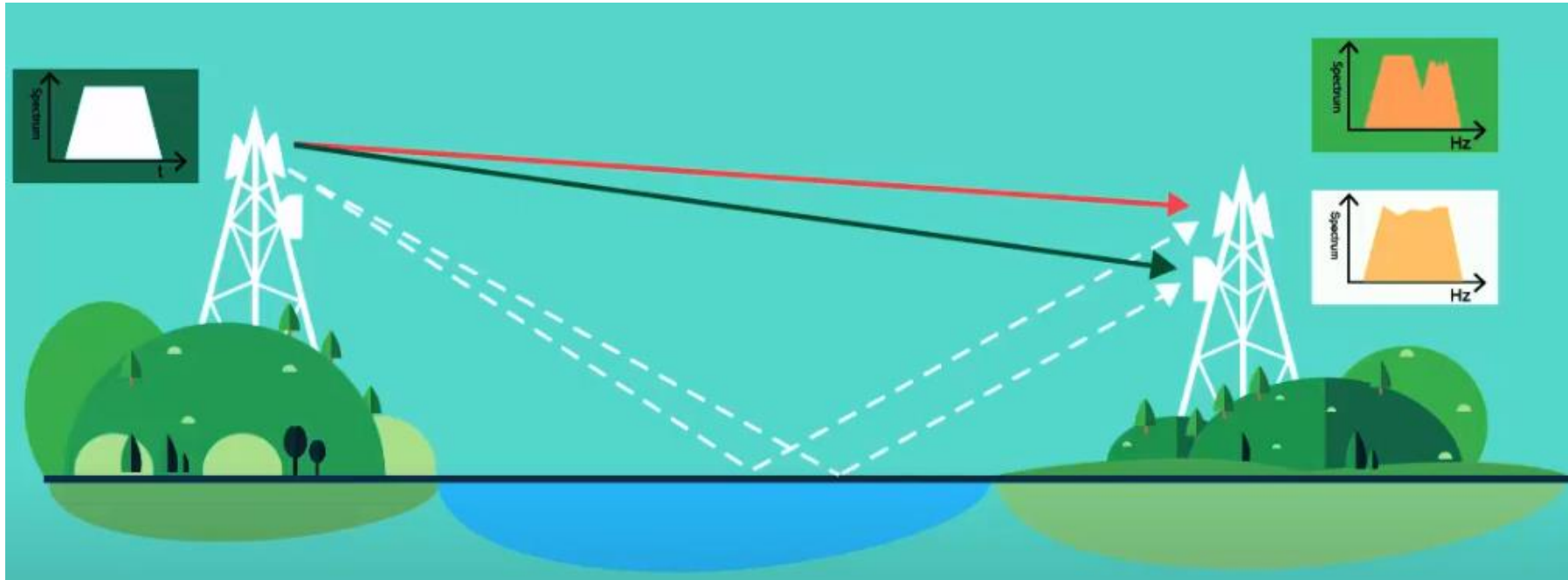


- ☑ Many phenomena require a single outcome to succeed. Most people are born with two **kidneys** and two **lungs**. If one of them **fails**, they can **still live** a healthy life **with some precaution**. **This is the essence of diversity.**

Diversity

Basic Concept

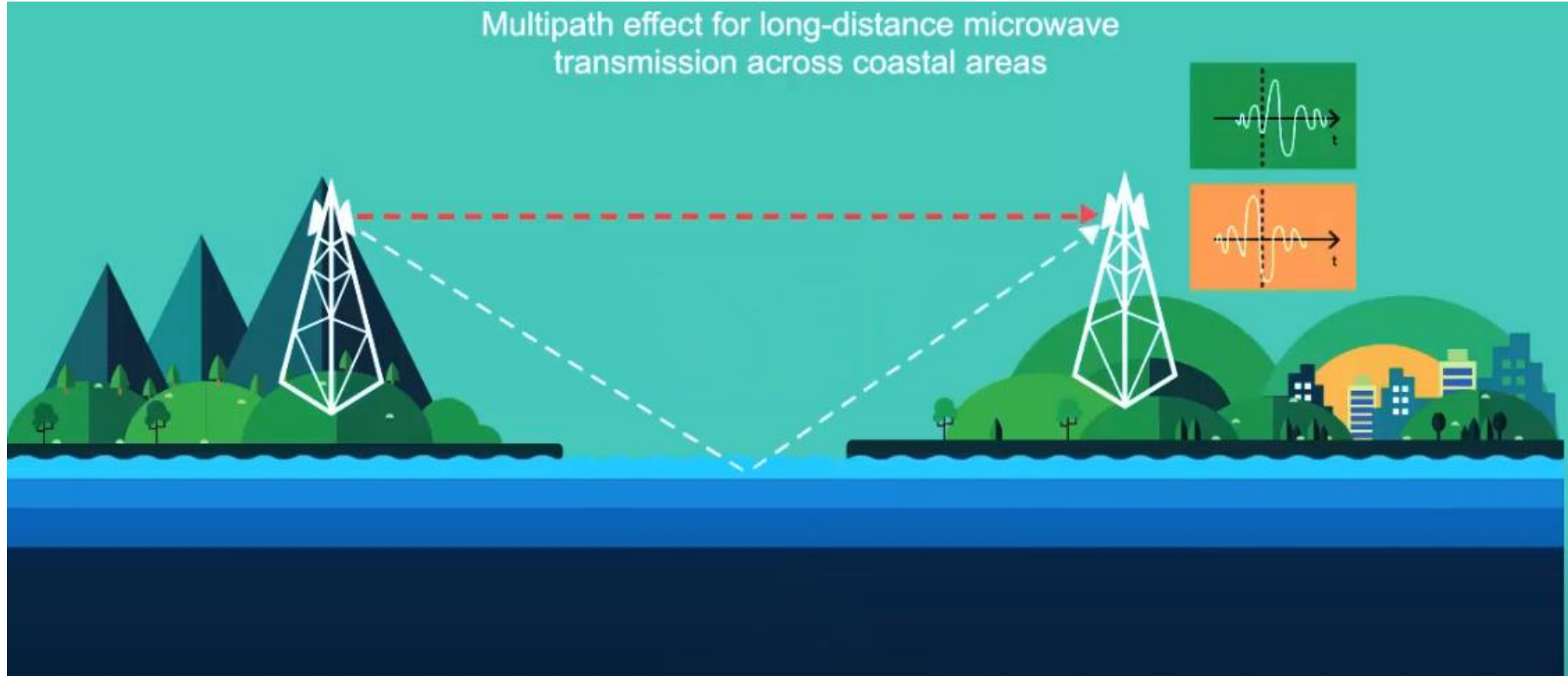
- ✓ Diversity uses multiple independent fading paths to transmit the same signal.
- ✓ Similar to increasing the chances of getting heads by tossing more coins.
- ✓ If any copy provides sufficient SNR, the system succeeds.
- ✓ A simple example is combining signals from two antennas to enhance reception.



Diversity

Multipath vs. Diversity Paths

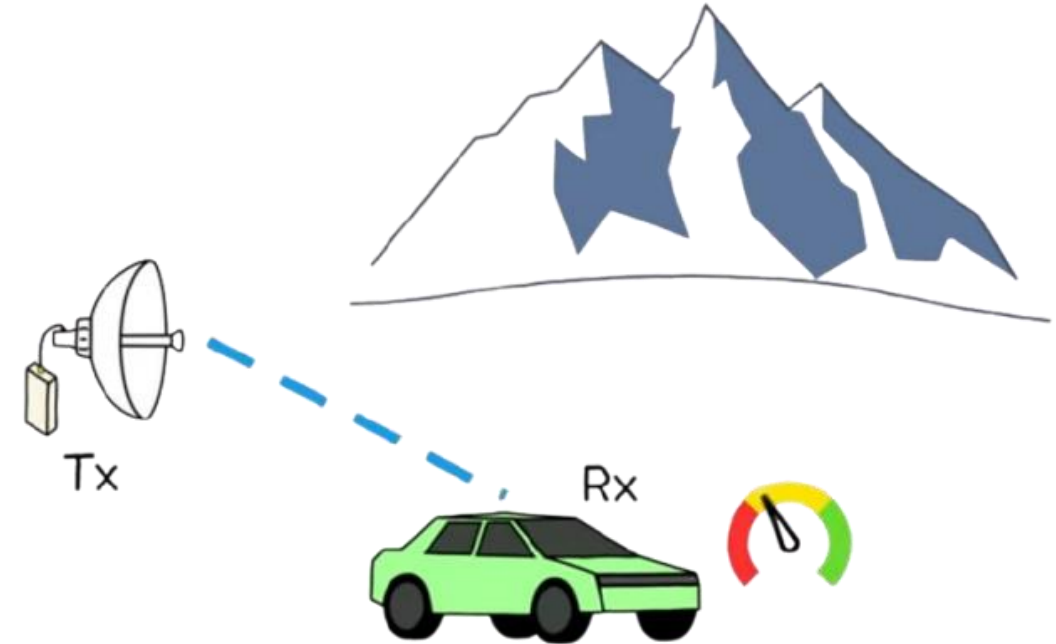
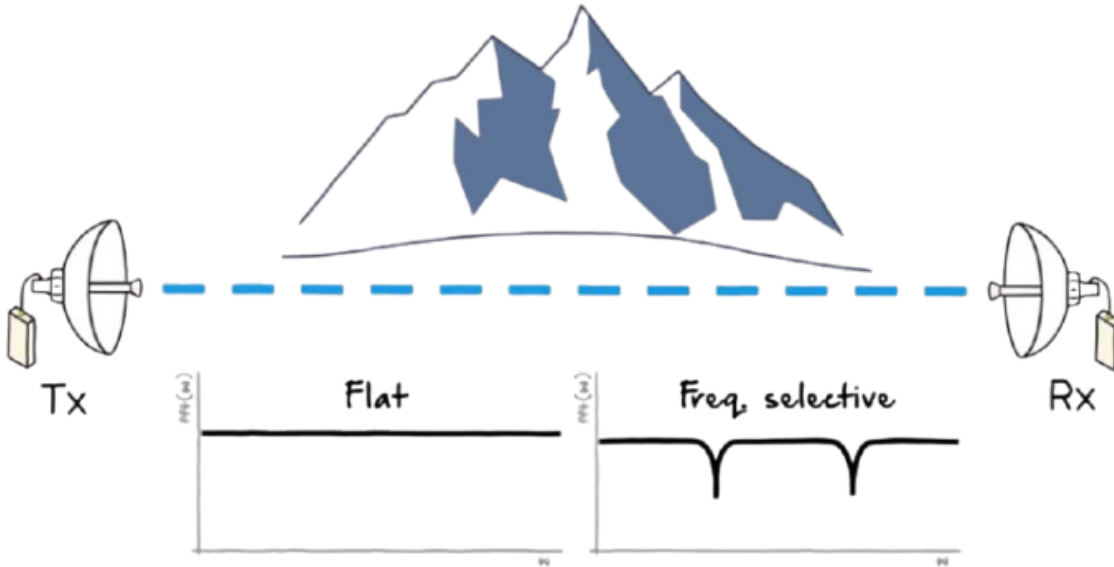
- ☑ **Multipath** effects are an **uncontrolled** phenomenon where signals **naturally travel** through **multiple paths** due to environmental reflections and diffraction.



Diversity

Multipath vs. Diversity Paths

☑ **Multipath** leads to flat, frequency selective fading or doppler shift fading.

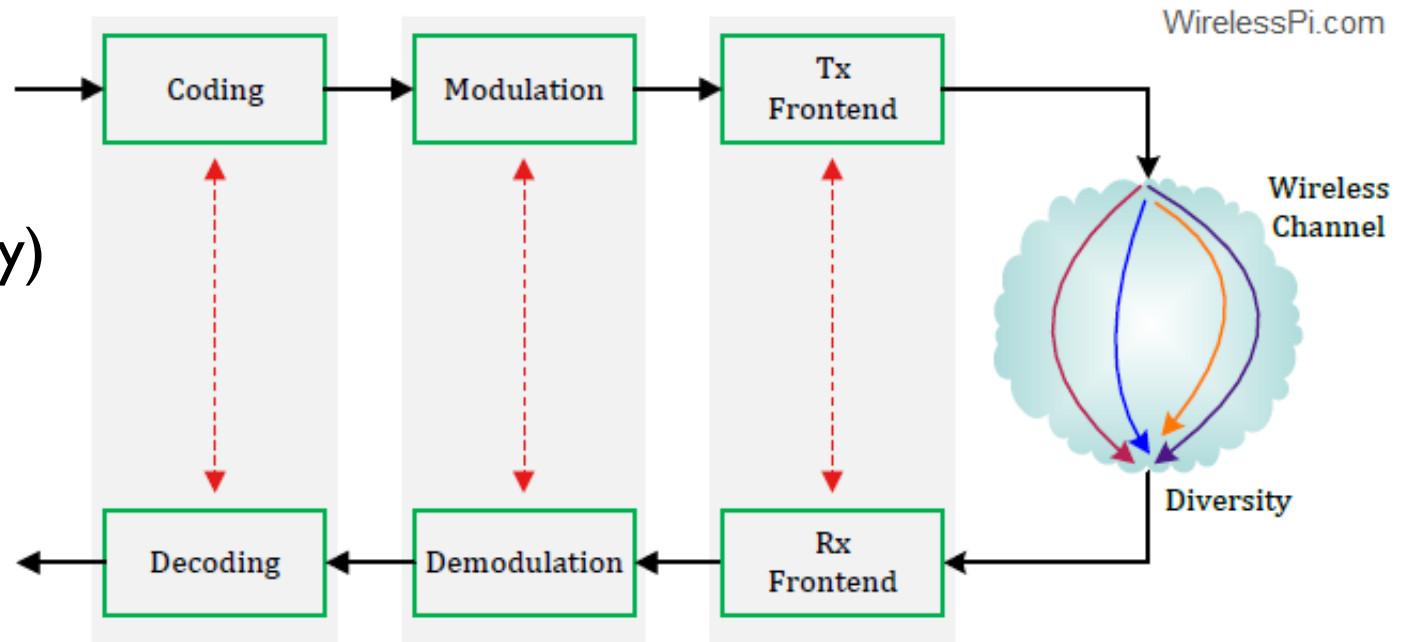


Diversity

Multipath vs. Diversity paths

☑ **Diversity paths (routes)** are **intentionally** designed techniques such as:

- ☛ Frequency diversity.
- ☛ Time diversity
- ☛ Spatial diversity (Antenna Diversity)
- ☛ Polarization diversity.



Diversity

Key Goals of Diversity in Wireless Communication

- 1) **Improve Signal Reliability** – Reduces the chances of deep fades and signal loss.
- 2) **Enhance Coverage** – Maintain connectivity even in challenging environments.
- 3) **Reduce Bit Error Rate (BER)** – Ensures more accurate data transmission.
- 4) **Increase Capacity & Throughput** – Enhances data rates and efficiency.
- 5) **Combat Multipath Fading** – Caused by reflections and obstacles.
- 6) **Improve Quality of Service (QoS)** – Improves user experience in voice, video, and data applications.

Diversity

From the Mathematical Perspective

If the probability of each copy being lost due to channel fading is P_A, P_B, P_C , and so on, the transmission will fail only if all copies fade simultaneously. This can be expressed as:

$$P(\text{failure}) = P_A \times P_B \times P_C \times \dots$$

If the probability of an individual fade given by $p \leq 1$ is the **same** for all the **replicas** (a realistic assumption), the probability of overall link failure with L diverse copies is given by

$$P(\text{failure}) = 1 - p^L$$

Diversity

From the Mathematical Perspective

- And the corresponding probability of success is given by

$$\begin{aligned} P(\text{success}) &= 1 - P(\text{failure}) \\ &= 1 - p^L \end{aligned}$$

- The higher the number of diverse replicas L , the lower the chances of a failed transmission. This number L is known as **diversity order**.
- This diversity enables 5G to overcome wireless challenges for high-speed transmission.

Diversity

Effect on BER

- Without proof, we state that in a Rayleigh fading channel, the deep fade probability p is inversely proportional to the SNR.

$$p \approx \frac{1}{SNR}$$

- The probability of failure or Bit Error Rate (BER) is related to SNR as.

$$BER \propto p^L = \frac{1}{SNR^L}$$

- An important insight from the expression:** higher diversity order reduces BER exponentially!

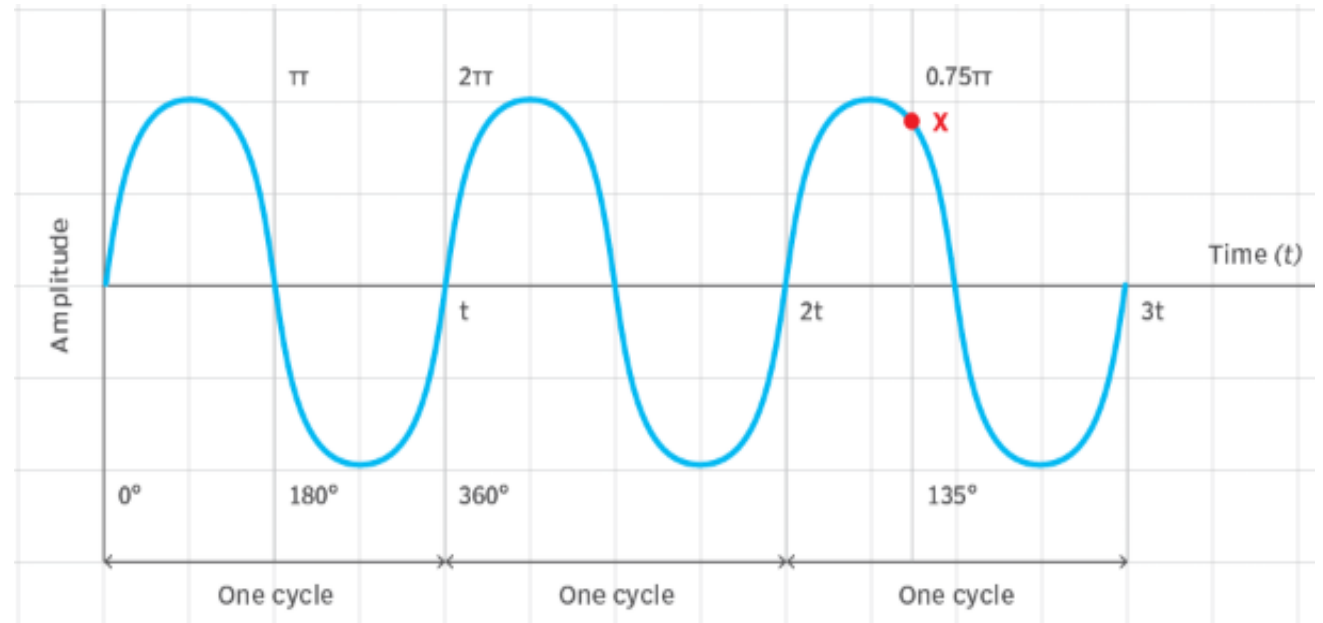
Diversity

☑ Phase, wavelength and path length

☑ **Multipath phenomena** cause **phase differences**, leading to destructive interference and signal fading.

☑ **Phase** refers to the **position of a wave within its cycle** at a given moment.

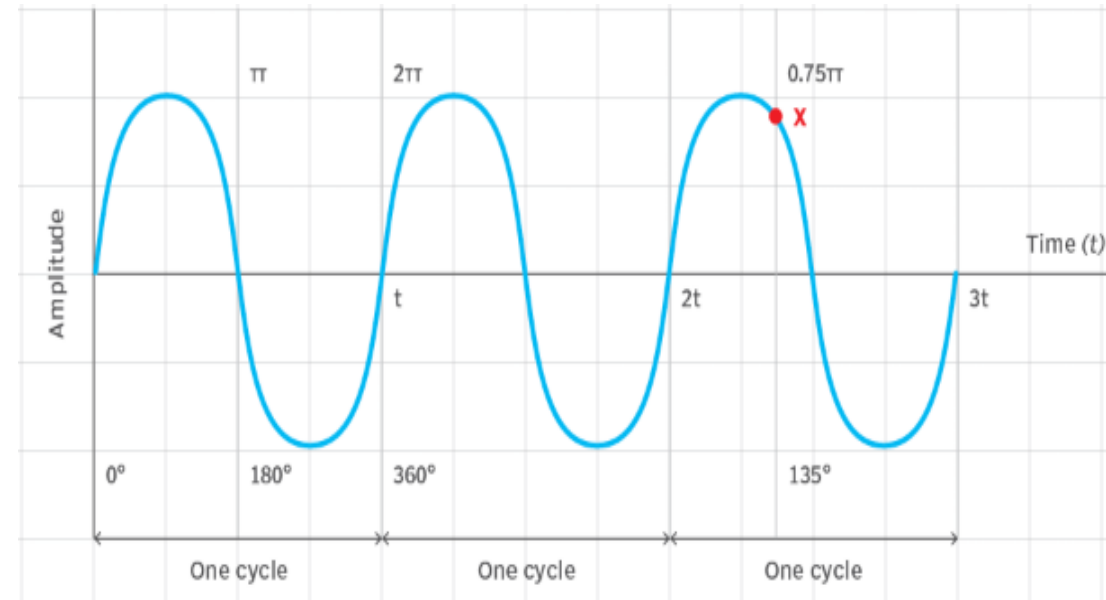
☑ A full wave cycle = 2π radians.



Diversity

☑ Phase, wavelength and path length

- ⇒ The wave starts at 0° (0π) with no amplitude
- ⇒ Peaks at 90° (0.5π radians)
- ⇒ Returns to rest at 180° (π radians)
- ⇒ Reaches a negative peak at 270° (1.5π radians)
- ⇒ Returns to rest at 360° (2π radians), beginning a new cycle.



Any point on the wave cycle can be identified by its phase. In the above Figure, point **X** in the third cycle represents a phase of about 135 degrees (0.75π radians).

Diversity

☑ Phase, wavelength and path length

Breaking It Down Step by Step:

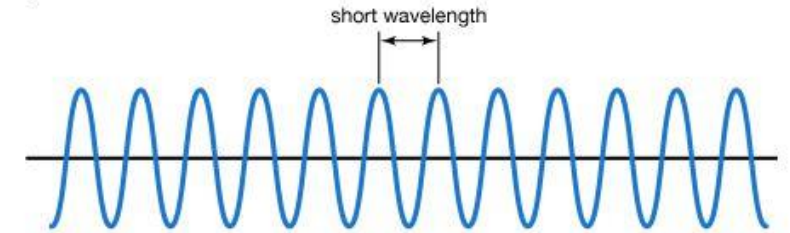
I. Wavelength & Path Length

1. A wave repeats itself every **wavelength (λ)**.
2. The total **path length (L)** determines how many full wavelengths fit in the path.

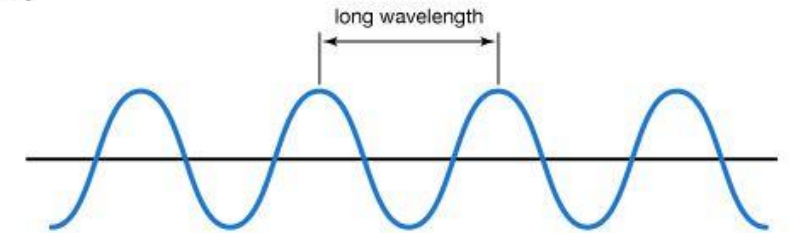
2. Counting Wavelengths in the Path

1. If the path length is an exact multiple of the wavelength (e.g., $L = 10\lambda$), the signal arrives **in-phase** (aligned).
2. If the path length includes a **fractional part** (e.g., $L = 10.5\lambda$), the signal arrives with a **phase shift**.

High frequency



Low frequency



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Diversity

☑ Phase, wavelength and path length

Breaking It Down Step by Step:

3. Phase Shift Due to Fractional Wavelengths

1. If the remaining fraction of a wavelength is **0.5λ** , the phase shift is **180°** (out of sync $\rightarrow$ destructive interference).
2. If the fraction is **0.25λ** , the phase shift is **90°** (partially misaligned).
3. If the fraction is **0λ or 1λ** , the phase shift is **0° or 360°** , meaning perfect alignment (constructive interference).

Diversity

☑ Phase, wavelength and path length

Breaking It Down Step by Step:

Example: Suppose a wave has a **wavelength** of **2** meters:

- 1) If the transmission path is **10** meters, then: $10/2=5$ full wavelengths .The wave arrives in **phase (aligned)**.
- 2) If the transmission path is **11 meters**, then: $11/2=5.5$ wavelengths.The extra **0.5λ** causes a **180° phase shift**, leading to **destructive interference**.

Diversity

☑ Phase, wavelength and path length

In general

- ☑ If we send a signal and it undergoes multipath interference, some frequency components may be severely attenuated due to phase cancellation.
- ☑ However, if we send another copy of the signal at a slightly different frequency, and this frequency difference is greater than the coherence bandwidth, the new signal will not experience the same destructive interference.
- ☑ This is because the phase difference between the two paths for the new frequency will not be exactly 180° , avoiding full cancellation.

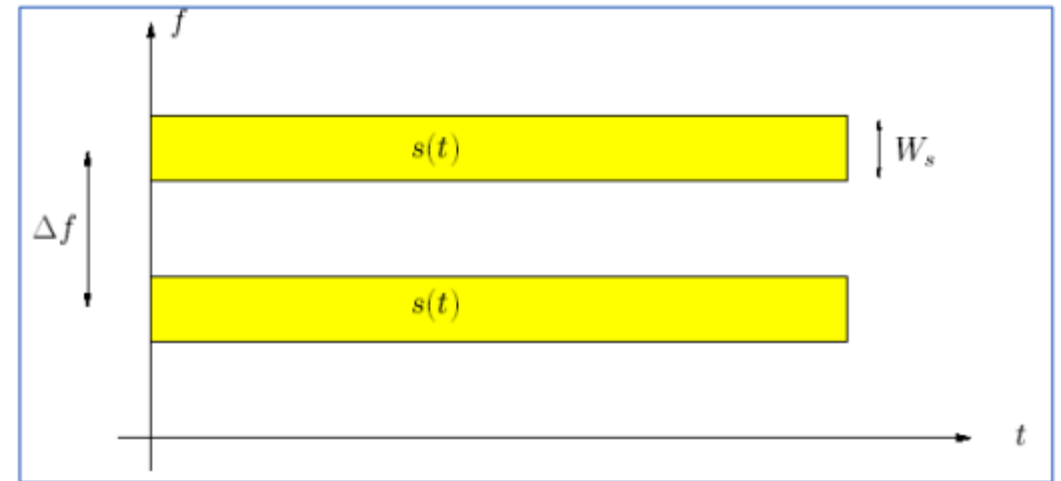
Diversity Types

☑ Frequency Diversity

The same information is transmitted over different carriers whose frequency **gap** Δf is greater than **coherence bandwidth** B_c of the channel.

- If $\Delta f > B_c$, the fading affecting one frequency is unlikely to affect the other, making frequency **diversity** effective.
- If $\Delta f < B_c$, both frequencies may experience similar fading, reducing the benefits of frequency diversity.
- Only one antenna is needed (both Tx & Rx)

Examples: Used in spread spectrum (FHSS, DSSS) and Orthogonal Frequency Division Multiplexing (OFDM).



Diversity Types

☑ Time Diversity

Time diversity improve signal reliability by transmitting **the same data multiple times** at different time intervals.

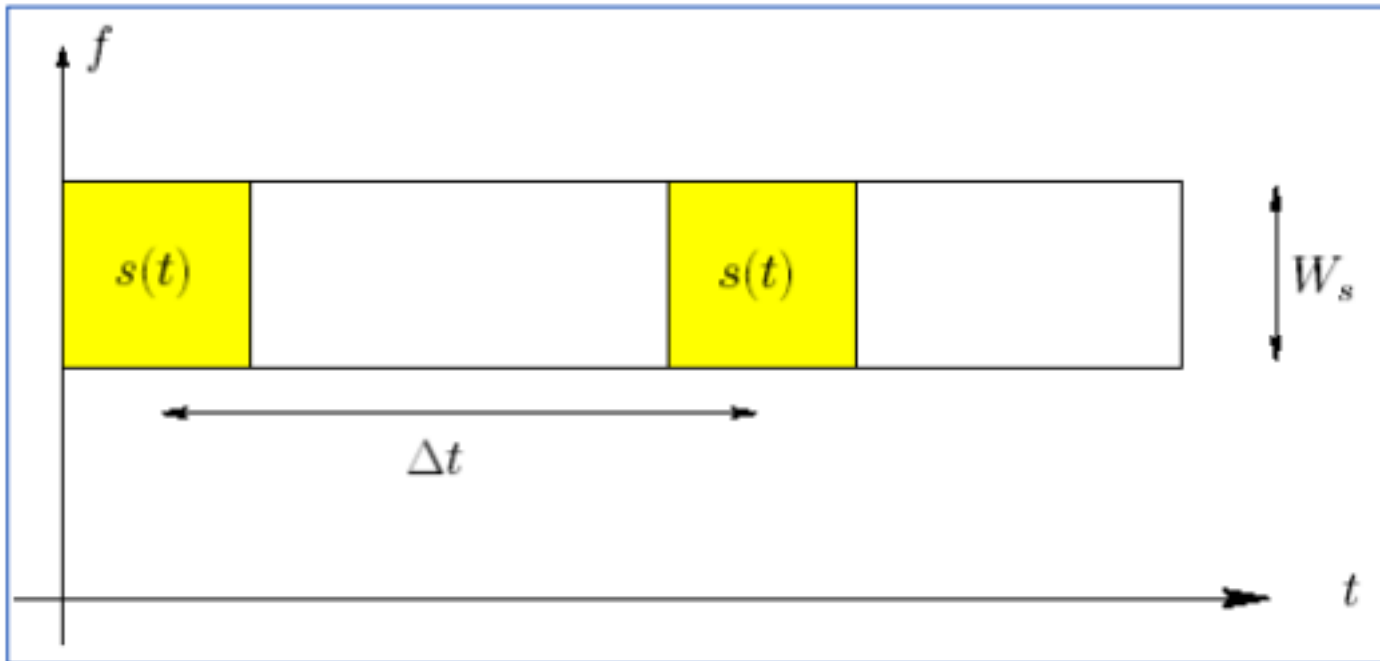
■ How Time Diversity Works:

- The key idea is that fading is time-dependent—if a signal experiences deep fading at one moment, it might be received clearly at another time.
- By retransmitting or interleaving data over time, the system ensures that at least some transmissions avoid deep fades.

Diversity Types

☑ Time Diversity: Relation to Coherence Time (T_c)

- ☐ Coherence Time (T_c) is the time duration over which the channel remains stable.
- ☐ For time diversity to be effective, the retransmissions must be spaced apart by a time interval Δt greater than T_c so that each transmission experiences independent fading conditions.
- ☐ If $\Delta t < T_c$, the channel conditions remain similar, reducing the effectiveness of time diversity.



Diversity Types

☑ Time Diversity

Techniques Used in Time Diversity:

- Interleaving:
 - Data bits are rearranged before transmission so that consecutive bits do not experience the same fade.
- Automatic Repeat reQuest (ARQ):
 - If a transmission fails, the system requests a retransmission at a later time when the channel might be better.
 - Used in TCP/IP protocols, LTE, and Wi-Fi.

Diversity Types

☑ Time Diversity

Techniques Used in Time Diversity:

- Time-Hopping Spread Spectrum (THSS):
 - The signal is transmitted at different time slots to avoid bursts of interference.
 - Used in ultra-wideband (UWB) communication.
- Multiple Time Slots in TDMA (Time Division Multiple Access):
 - Users are assigned different time slots, reducing the chance of all transmissions being affected by deep fading at the same moment.

Diversity Types

■ Space Diversity

- ⇒ **Space diversity** (also called **spatial diversity**) is a technique used to improve signal reliability by using **multiple antennas** at **different locations** to transmit or receive the same signal.
- ⇒ known as **Multiple-Input Multiple-Output (MIMO)**
- ⇒ This helps combat fading and interference caused by multipath propagation.

Diversity Types

■ Space Diversity

○ How Space Diversity Works:

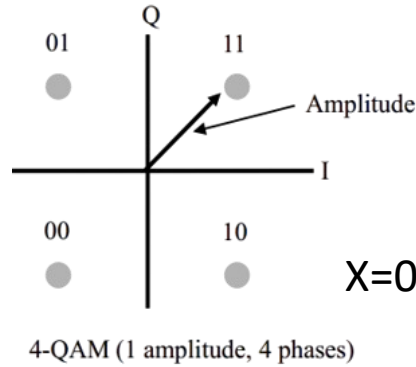
- When signals travel through **different paths**, some may **experience deep fading** while others **remain strong**.
- By using multiple antennas spaced apart, at least one antenna is likely to receive a strong signal.
- The receiver then selects the best signal or combines them to improve performance.

Diversity Types

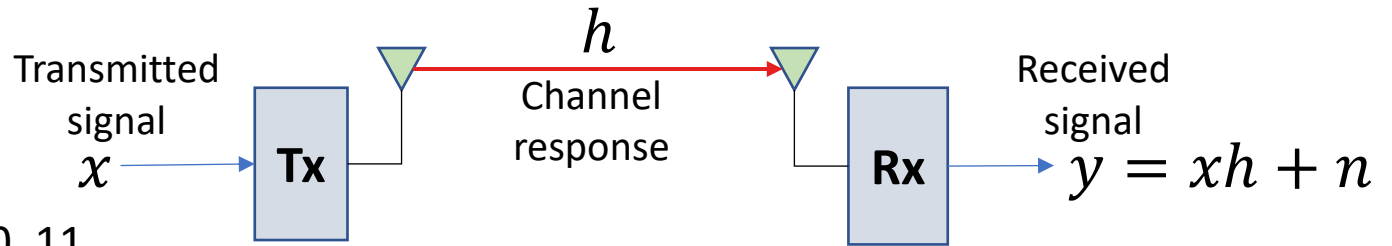
■ Space Diversity: Single input Single output (SISO)



SISO



$X=00, 01, 10, 11$



$$y = xh + n$$

- ✓ The coefficient h characterizes the effect of the channel on the transmitted signal.
- ✓ h is modelled as a complex random variable (RV) due to multipath fading.
- ✓ Its mean and variance depend on the statistical fading model used, such as:
 - Rayleigh fading or Rician fading.
- ✓ x is modelled as RV.

■ Single input Single output (SISO)

Instantaneous SNR Definition

The **instantaneous SNR** at the receiver is defined as the ratio of the received signal power to the noise power:

$$\gamma = \frac{\text{Received Signal Power}}{\text{Noise Power}}$$

The received signal power depends on the channel gain $|h|^2$, so:

$$\gamma = \frac{P_s |h|^2}{P_n}$$

where:

- $P_s = E[|x|^2]$ is the **transmitted signal power**.
- $P_n = E[|n|^2]$ is the **noise power**.
- $|h|^2$ is the **channel power gain**, which follows an exponential distribution in Rayleigh fading.

If $|h|^2 = 1$ then channel has no effect on the signal power. In such case:

$$SNR = \frac{P_s}{P_n}$$

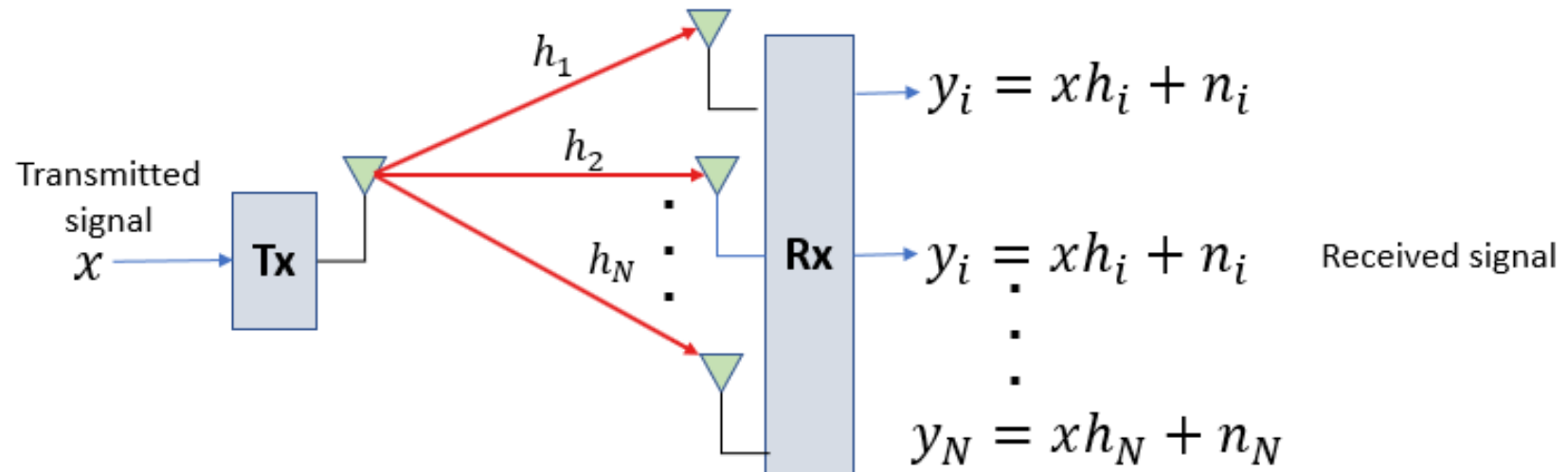
■ Single Input Multiple Output (SIMO)



✓ SIMO is a communication system with one transmit antenna and multiple receive antennas.

✓ Also known as **receive diversity**, one of the oldest applications of multiple antennas.

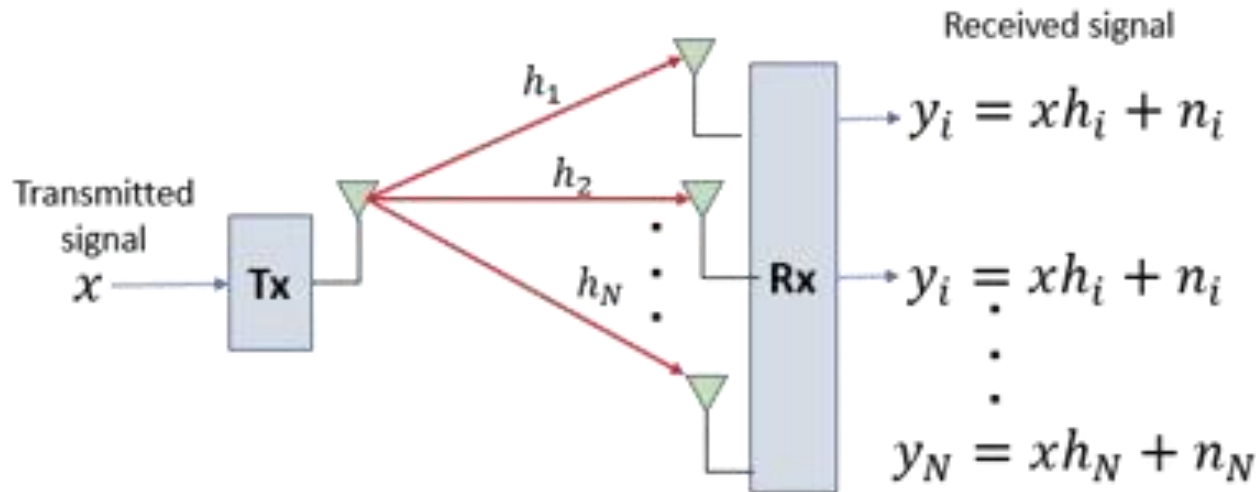
✓ Involves combining outputs from multiple receive antennas using analog or digital processing..



■ Single Input Multiple Output (SIMO)

Mathematical Model

For each receive antenna i , the received signal y_i can be written as:



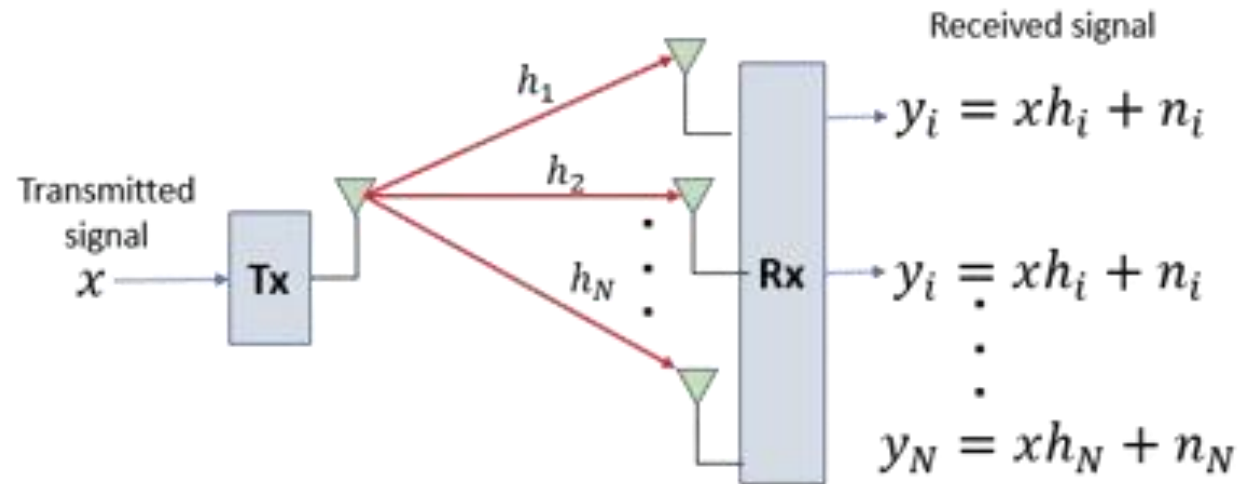
$$y_i(t) = x(t)h_i(t) + n_i(t)$$

Where:

- $x(t)$ is the transmitted signal,
- $h_i(t)$ is the channel gain between the transmitter and the i -th receive antenna,
- $n_i(t)$ is the additive white Gaussian noise (AWGN) at the i -th receive antenna.

■ Single Input Multiple Output (SIMO)

Mathematical Model



Thus, the received signal vector at all receive antennas can be expressed as:

$$\mathbf{y}(t) = \mathbf{x}(t)\mathbf{H}(t) + \mathbf{n}(t)$$

$$\mathbf{y}(t) = [y_1(t), y_2(t), y_3(t), \dots, y_N(t)]^T$$

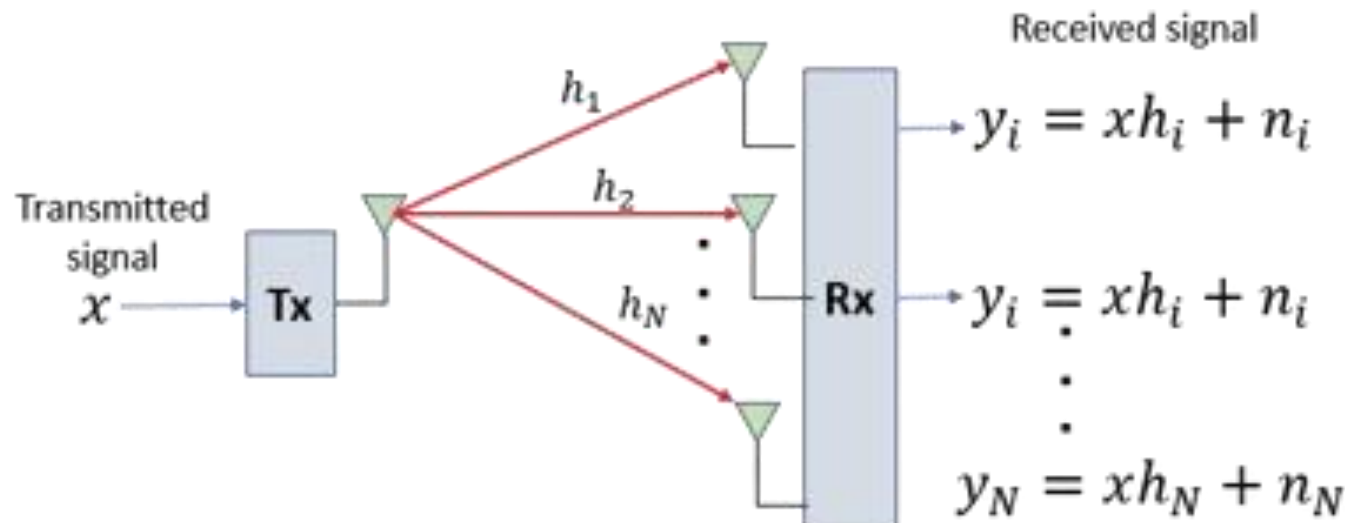
$$\mathbf{n}(t) = [n_1(t), n_2(t), \dots, n_N(t)]^T.$$

$\mathbf{H}$ is the matrix of channel gains: $\mathbf{H} = \begin{bmatrix} h_1(t) \\ h_2(t) \\ \vdots \\ h_N(t) \end{bmatrix}$

■ Single Input Multiple Output (SIMO)

Combining techniques: Selection Combining (SC)

- ✓ The **receiver selects the antenna** with the **highest received signal strength**.
- ✓ The **chosen antenna's signal** is used for further processing.
- ✓ **Advantages**: Simple to implement and effective in combating fading.
- ✓ **Disadvantages**: Does **not provide significant diversity gain** since only one antenna is used at a time.



Selected antenna

$$y = \max_{\forall i} |y_i|$$

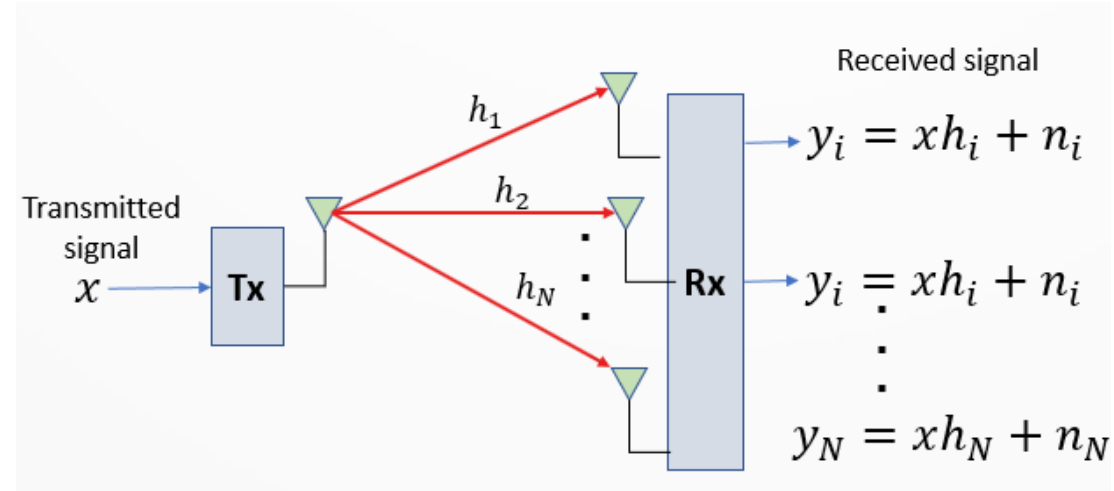
The receiver selects the antenna with the highest instantaneous SNR

$$SNR = \max_{i \in \{1, 2, 3, \dots, N\}} \frac{|h_i|^2 P}{N_0}$$

■ Single Input Multiple Output (SIMO)

Combining techniques: Maximum Ratio Combining (MRC)

- ✓ The received signals from all antennas are **weighted** and **combined** based on their SNR to maximize the received signal power.
- ✓ The end result is a reconstructed signal with an improved SNR and receiver sensitivity
- ✓ The final combined MRC is given by:



$$y = \sum_{i=1}^N w_i y_i \quad \text{The optimal weights are:} \quad w_i = \frac{h_i^*}{|h_i|^2}$$

where:

→ h_i^* = Complex **conjugate** of the channel gain h_i

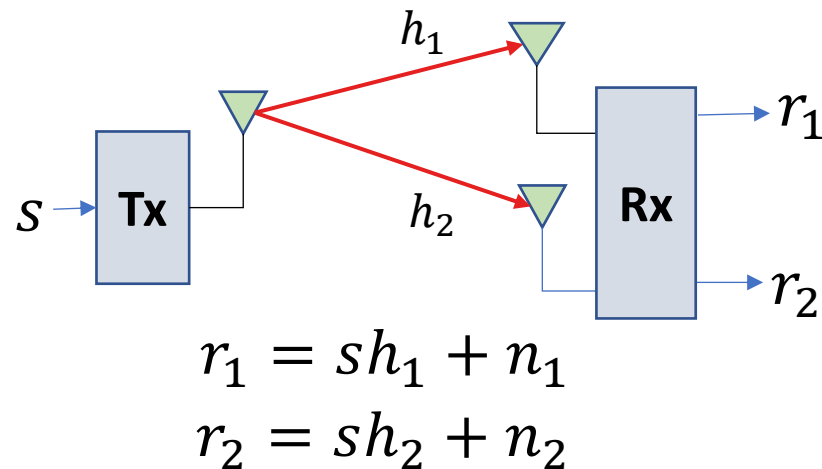
→ $|h_i|^2$ = Power of the channel gain.

$$\text{SNR}_{\text{MRC}} = \sum_{i=1}^N \frac{|h_i|^2 P}{N_0}$$

■ Single Input Multiple Output (SIMO)

☑ Maximum Ratio Combining (MRC): Example

Let's consider a **SIMO** with **two receiving antennas** ($N = 2$).



Step 1: Given System Parameters

We assume:

- Transmitted signal: $s = 1$
- Channel gains:
 - $h_1 = 0.8 + 0.6j$
 - $h_2 = 0.5 - 0.3j$
- Noise at each antenna:
 - $n_1 = 0.2 - 0.1j$
 - $n_2 = 0.1 + 0.2j$

Each receiving antenna gets a different version of the signal due to fading and noise.

■ Single Input Multiple Output (SIMO)

☑ Maximum Ratio Combining (MRC): Example

Step 2: Received Signals

Each antenna receives a signal given by:

$$r_i = h_i s + n_i$$

For **Antenna 1**:

$$r_1 = (0.8 + 0.6j) \cdot 1 + (0.2 - 0.1j)$$

$$r_1 = (0.8 + 0.6j) + (0.2 - 0.1j)$$

$$r_1 = 1.0 + 0.5j$$

For **Antenna 2**:

$$r_2 = (0.5 - 0.3j) \cdot 1 + (0.1 + 0.2j)$$

$$r_2 = (0.5 - 0.3j) + (0.1 + 0.2j)$$

$$r_2 = 0.6 - 0.1j$$

■ Single Input Multiple Output (SIMO)

☑ Maximum Ratio Combining (MRC): Example

Step 3: Compute Weights w_i

The MRC weights are given by:

$$w_i = \frac{h_i^*}{|h_i|^2}$$

For Antenna 1:

- Compute magnitude squared of h_1 :

$$|h_1|^2 = (0.8)^2 + (0.6)^2 = 0.64 + 0.36 = 1.0$$

- Compute complex conjugate of h_1 :

$$h_1^* = 0.8 - 0.6j$$

- Compute weight:

$$w_1 = \frac{0.8 - 0.6j}{1.0} = 0.8 - 0.6j$$

■ Single Input Multiple Output (SIMO)

☑ Maximum Ratio Combining (MRC): Example

For Antenna 2:

- Compute **magnitude squared** of h_2 :

$$|h_2|^2 = (0.5)^2 + (-0.3)^2 = 0.25 + 0.09 = 0.34$$

- Compute **complex conjugate** of h_2 :

$$h_2^* = 0.5 + 0.3j$$

- Compute **weight**:

$$w_2 = \frac{0.5 + 0.3j}{0.34} = 1.47 + 0.88j$$

■ Single Input Multiple Output (SIMO)

☑ Maximum Ratio Combining (MRC): Example

Step 4: Compute Combined Signal y

MRC output is:

$$y = w_1 r_1 + w_2 r_2$$

Compute $w_1 r_1$:

$$(0.8 - 0.6j)(1.0 + 0.5j)$$

Using the distributive property:

$$= (0.8 \cdot 1.0) + (0.8 \cdot 0.5j) + (-0.6j \cdot 1.0) + (-0.6j \cdot 0.5j)$$

$$= 0.8 + 0.4j - 0.6j + 0.3$$

$$= 1.1 - 0.2j$$

■ Single Input Multiple Output (SIMO)

☑ Maximum Ratio Combining (MRC): Example

Compute $w_2 r_2$:

$$(1.47 + 0.88j)(0.6 - 0.1j)$$

Expanding:

$$= (1.47 \cdot 0.6) + (1.47 \cdot -0.1j) + (0.88j \cdot 0.6) + (0.88j \cdot -0.1j)$$

$$= 0.882 - 0.147j + 0.528j - 0.088$$

$$= 0.794 + 0.381j$$

Compute final output y :

$$y = (1.1 - 0.2j) + (0.794 + 0.381j)$$

$$y = 1.894 + 0.181j$$

■ Single Input Multiple Output (SIMO)

☑ Maximum Ratio Combining (MRC): Example

Final Result

The final output signal after **MRC combining** is:

$$y = 1.894 + 0.181j$$

This is a **stronger and more reliable signal** compared to the original received signals $r_1 = 1.0 + 0.5j$ and $r_2 = 0.6 - 0.1j$. The **noise effect is reduced**, and the **signal power is maximized**.

■ Single Input Multiple Output (SIMO)

☑ Maximum Ratio Combining (MRC): Example

Step 1: Compute Individual SNRs

For Antenna 1:

$$|h_1|^2 = (0.8)^2 + (0.6)^2 = 0.64 + 0.36 = 1.0$$

For Antenna 2:

$$|h_2|^2 = (0.5)^2 + (-0.3)^2 = 0.25 + 0.09 = 0.34$$

Assume Noise Power $N_0 = 0.1$ (example value):

$$\text{SNR}_1 = \frac{1.0}{0.1} = 10$$

$$\text{SNR}_2 = \frac{0.34}{0.1} = 3.4$$

■ Single Input Multiple Output (SIMO)

☑ Maximum Ratio Combining (MRC): Example

Step 2: Compute Total SNR after MRC

$$\text{SNR}_{\text{MRC}} = \text{SNR}_1 + \text{SNR}_2$$

$$\text{SNR}_{\text{MRC}} = 10 + 3.4 = 13.4$$

Final Answer

The SNR after MRC is:

$$\text{SNR}_{\text{MRC}} = 13.4$$

This is higher than the individual SNRs, confirming that MRC improves SNR by combining signals optimally.

THANKS

